Project Report

On

Student Management System

(BCA-605)

Submitted By:

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BCA 6th Sem

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M.M. Institute of Computer Technology & Business Management

MAHARISHI MARKANDESHWAR (DEEMED TO BE UNIVERSITY)

(NAAC accredited Grade ‘A++’University)

DECLARATION

I hereby declare that the work which is being presented in this report entitled **"STUDENT MANAGEMENT SYSTEM",** in partial fulfillment of the requirement for the award of the degree of **BACHELOR OF COMPUTER APPLICATIONS** submitted at M.M. Institute of Computer Technology & Business Management, **Maharishi Markandeshwar (Deemed to be University), Mullana, Ambala** is an authentic work under The Guidance of "**Dr. Varun Gupta**".

The work presented in this report not been submitted by me for the award of any other degree Of this or any other Institute/University.

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Date: ____________________

This is certify that the above statement made by the candidate is correct to the best of my Knowledge and belief.

Signature
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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the successful completion of my project titled **“Student Management System”**

First and foremost, I extend my heartfelt thanks to my **project guide, [Dr. Varun Gupta]**, for their constant guidance, valuable suggestions, and continuous encouragement during every stage of this project. Their support and motivation have been instrumental in completing this work successfully.

I am also thankful to the **Head of Department** and all the **faculty members** of the Department of Computer Science for providing the necessary facilities, technical knowledge, and academic assistance that made this project possible.

A special thanks to my **friends and classmates** for their cooperation, feedback, and moral support throughout the development of the system.

Lastly, I would like to express my deepest gratitude to my **family** for their constant love, patience, and encouragement during this project.

ABSTRACT

Student Management System is a **web-based platform** designed to digitize the traditional **notice board** used in **schools, colleges, offices, and institutions**. In the modern era, the timely and efficient dissemination of **information** is critical for organizational functioning. Traditional **notice boards**, which rely on **manual posting** and **physical updates**, are often **time-consuming, inefficient**, and prone to **misplacement or delay**, leading to poor communication. The **Student Management System** addresses these limitations by allowing **administrators** to **create, manage, and display notices electronically**, while enabling **users** to **access information securely and conveniently** from any device. The system is implemented using **PHP** for **server-side scripting**, **MySQL** for **database management**, and **HTML** and **CSS** for the **frontend interface**, with **XAMPP** serving as the **development and hosting environment**.

The system incorporates a **secure login and registration module**, which ensures that **users** can create accounts and **authenticate** before accessing the platform. This functionality guarantees that **only authorized users** can view notices, while **administrators** have exclusive rights to manage content. The **admin panel** provides comprehensive control, allowing **creating, editing, deleting, and scheduling notices**, ensuring that all updates are timely and accurate. **PHP sessions** and **database authentication** protect sensitive administrative features, preventing **unauthorized access** and maintaining **data integrity**.

The **user interface** is designed to be **responsive, intuitive, and visually appealing**, allowing **users** to **view notices in chronological order**, **search for specific information**, and **filter notices by date or category**. The **MySQL database** efficiently stores **user information, login credentials, and notice details** such as titles, descriptions, dates, and categories. By replacing **manual notice boards** with a **digital system**, the platform eliminates **physical posting**, reduces **paper usage**, and ensures **instant access** to important **announcements**, significantly improving **communication efficiency** within the organization.

Moreover, the **E-Notice Board System** supports **real-time updates**, enabling **users** to be immediately notified of new notices. The platform is **device-independent**, accessible via **desktops, laptops, tablets, or smartphones**, making it highly adaptable for modern organizational needs. Future enhancements could include **email notifications, document attachments, multimedia support, and multi-level access control**, further extending its functionality. By integrating **PHP, MySQL, HTML, CSS, and js**, the system provides a **secure, scalable, and reliable platform** for managing notices digitally. Overall,

the **E-Notice Board System** represents a **modern, efficient, and eco-friendly solution** for organizations seeking to streamline **information dissemination** and enhance **administrative productivity**.

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INTRODUCTION

Airline passenger reviews play a crucial role in shaping the reputation and success of airlines in today's competitive market. With thousands of passengers sharing their experiences online, airlines have access to a massive amount of feedback that reflects customer satisfaction, expectations, and pain points. However, manually analyzing this data is time-consuming and often fails to provide meaningful insights**.**

To overcome this challenge, sentiment analysis—a Natural Language Processing (NLP) technique—is used to automatically identify whether a review expresses a positive, negative, or neutral opinion. This project, ***"Sentiment Analysis on Airline Passenger Reviews,"*** leverages Python libraries such as **NumPy** and **Pandas** for data cleaning, preprocessing, and analysis, while **Power BI** is used for creating interactive visual dashboards.

The project aims to transform raw, unstructured textual data into valuable business insights. Through this analysis, airlines can understand passengers' sentiments toward various aspects of service such as punctuality, crew behavior, food quality, and comfort level. Ultimately, this data-driven approach helps airlines improve service quality, enhance customer satisfaction, and make informed business decisions based on factual evidence rather than assumptions.

1.1 Objectives

- To collect and analyze airline passenger reviews to understand customer sentiments.
- To preprocess and clean textual data using Python libraries like NumPy and Pandas.
- To classify reviews into **positive, negative, or neutral** categories based on sentiment.
- To visualize sentiment trends and patterns using **Power BI dashboards**.
- To identify key factors influencing customer satisfaction and dissatisfaction.
- To support airline management in **data-driven decision-making** for service improvement.

1.2 Scope:

The scope of this project focuses on analyzing public airline passenger reviews to evaluate customer satisfaction levels using **data analytics and visualization tools**. The dataset contains various attributes such as airline name, review text, rating, and flight class. The scope extends from **data collection and cleaning** to **sentiment categorization and visualization**. This project applies **Python (NumPy and Pandas)** for handling large datasets and preprocessing text data, ensuring that the data is accurate, structured, and ready for analysis.

1.3 Existing system

In the existing system, most airlines rely heavily on **manual review monitoring** or **basic rating averages** from passengers to assess service quality. Customer feedback is often collected through surveys, email responses, or social media posts, but these reviews are rarely analyzed in depth due to the large volume of data.

Manual analysis of customer reviews is time-consuming, inconsistent, and lacks the ability to identify trends or patterns effectively. Existing systems often focus only on numeric ratings and ignore textual feedback, which contains valuable insights about passenger emotions, preferences, and experiences.

1.4 Proposed system

The proposed system introduces a **data-driven and automated approach** for analyzing airline passenger reviews through **sentiment analysis and visualization tools**. The system uses **Python libraries (NumPy, Pandas)** to process, clean, and analyze the data efficiently. Reviews are categorized as **positive, negative, or neutral** based on textual content and sentiment polarity.

After processing, the insights are represented visually using **Power BI dashboards**, allowing users to interactively explore key metrics such as sentiment distribution by airline, flight class, and overall satisfaction trends. This system eliminates the limitations of manual analysis by offering **faster, more accurate, and visually clear** results.

FEASIBILITY STUDY

Generally, feasibility studies precede technical development and project implementation. Traditionally uncover the strengths and weaknesses of the existing system or proposed venture. In its simplest term, the two criteria to judge feasibility are cost required and value to be attained. As such, a well-designed feasibility study should provide historical background of the project assessment of feasibility study is based on the following factors:

1) Technical Feasibility

2) Economic Feasibility

3) Operational Feasibility

2.1 Technical Feasibility:

The Student Management System is technically feasible to develop using basic web technologies such as **HTML, CSS, and JavaScript**, as these technologies are capable of building responsive user interfaces, form-driven modules, and interactive dashboards needed for student data handling. HTML and CSS provide the structure and design for pages like admissions, attendance, fees, and reports, while JavaScript enables client-side validation, dynamic content updates, and interaction with backend APIs

2.2 Economic Feasibility

The economic feasibility of the sentiment analysis project on airline passenger reviews is highly favorable, as it can be implemented with minimal cost and substantial potential benefits.:

Cost-Saving Strategies

- **Use of Open-Source Tools:** The project relies on free and open-source technologies such as Python, NLTK, TextBlob, Scikit-learn, and Pandas, eliminating the need for paid software licenses.
- **Utilization of Free Datasets:** Publicly available airline review datasets are used for training and testing, avoiding expenses related to data collection.
- **Cloud-Based or Local Deployment:** The system can be deployed on low-cost cloud platforms like Google Colab or existing local hardware, reducing infrastructure costs.

2.3 Operational Feasibility:

The Student Management System is operationally feasible because it fits well into the daily workflows of administrators, teachers, students, and parents. The system centralizes all student-related processes—such as admissions, attendance, grades, fees, and communication—making them easier and faster to manage compared to manual registers or spreadsheets. Staff members can quickly learn to use the system because the interface is simple and built with familiar web technologies like HTML, CSS, and JavaScript, requiring minimal technical training.

2.4 Conclusion of Feasibility Study:

The feasibility study clearly shows that developing a Student Management System is practical, beneficial, and sustainable for the institution. From a **technical perspective**, the system can be effectively built using HTML, CSS, and JavaScript along with a simple backend, ensuring compatibility, ease of development, and future scalability. The **operational analysis** confirms that the system fits smoothly into existing workflows, reduces manual effort, minimizes errors, and improves communication among administrators, teachers, students, and parents. The **economic evaluation** indicates that the project is cost-effective, offering significant savings in time, resources, and administrative workload compared to manual methods. Overall, the system is feasible across all dimensions and will strengthen institutional management, enhance productivity, and support better decision-making. Therefore, the Student Management System is recommended for implementation as it promises high efficiency, reliability, and long-term value..

TOOLS & TECHNOLOGIES USED:

3.1 Hardware Requirements

- **Processor:** Intel Core i3 or equivalent (2.0 GHz or higher)
- **RAM:** 4 GB
- **Hard Disk:** 250 GB (for project files, datasets, and libraries)

3.2 Software Requirements

The Student Management System requires a web-based software environment built using HTML, CSS, and JavaScript for the frontend, along with a lightweight backend such as Node.js/Express or a serverless cloud database like Firebase to manage authentication and data storage. The system must support secure login, role-based access for admins, teachers, students, and parents, and provide modules for admissions, student profiles, attendance, grades, fees, and communication. It should allow users to perform tasks such as adding and updating student records, marking attendance, entering marks, generating report cards, managing fee payments, and sending notifications.

3.3 Technologies Used:

The Student Management System is developed using **HTML, CSS, and JavaScript**, which together form the core technologies of modern web development. **HTML (HyperText Markup Language)** is used to design the structure of the system, including pages for student registration, attendance, grades, fees, and reports. It defines the layout of forms, tables, buttons, headings, and all visual components. **CSS (Cascading Style Sheets)** is used to style and beautify the interface, making the system user-friendly and visually appealing. It handles colors, fonts, spacing, alignment, responsiveness, and overall design to ensure the application works smoothly on both desktop and mobile devices. **JavaScript** is the main programming logic of the system, enabling interactivity such as form validation, dynamic content updates, searching, filtering, and communicating with backend services through APIs. It allows real-time actions like marking attendance, calculating grades, updating dashboards, or displaying notifications without reloading the page. Together, HTML, CSS, and JavaScript provide a complete environment to build a responsive, interactive, and easy-to-use Student Management System..

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TOOLS DISCUSSION

he Student Management System is built using standard web technologies—**HTML, CSS, and JavaScript**—which together provide a simple, flexible, and efficient platform for developing interactive web applications. **HTML** serves as the foundational markup language used to structure all pages of the system, such as student profiles, attendance forms, dashboards, and reports. It defines the layout and organizes content so that it can be easily displayed and accessed by users. **CSS** is used to enhance the presentation of the system by applying styles, colors, layouts, and responsive designs. This ensures that the interface remains visually appealing, easy to navigate, and compatible with different screen sizes like mobile phones, tablets, and desktops. **JavaScript** adds dynamic functionality to the system by handling user interactions, validating form inputs, updating data without refreshing the page, and communicating with backend services through APIs or serverless functions. JavaScript also enables instant operations such as searching students, filtering records, calculating grades, showing notifications, and generating real-time updates on the dashboard.

Together, HTML, CSS, and JavaScript provide a lightweight, browser-friendly, and widely supported technology stack that makes the Student Management System fast, intuitive, and easy to maintain. These technologies require no installation on the client side, work across all major browsers, and offer enough flexibility to integrate future features like authentication, cloud storage, or payment gateways. Hence, this combination of technologies is well-suited for developing a simple yet functional student management application.

About the Tools

1. Visual Studio Code (VS Code)

- A powerful and lightweight code editor used to write HTML, CSS, and JavaScript.
- Offers features like syntax highlighting, IntelliSense (auto-completion), debugging, and an integrated terminal.
- Supports thousands of extensions such as Live Server, Prettier, and GitHub integration, which make development faster and easier.
- Allows real-time preview of the project through extensions and helps maintain clean, readable code.

2. Web Browsers (Google Chrome, Mozilla Firefox, Microsoft Edge)

- Browsers are used to run and test the Student Management System.
- Each browser comes with Developer Tools that allow you to inspect HTML elements, modify CSS, debug JavaScript, and monitor network requests.
- Helps check responsiveness on different screen sizes using device emulation.

- Useful for performance testing and identifying layout or script errors quickly

1. HTML (HyperText Markup Language)

- HTML is the **basic building block** of the Student Management System.
- It is used to create the **structure** of all web pages such as login page, dashboard, student registration form, attendance page, and reports page.
- Defines elements like headings, tables, forms, buttons, images, and links.
- Helps organize student data in structured formats using lists, tables, and div sections.
- Works as the skeleton of the entire project on which CSS and JavaScript operate.

2. CSS (Cascading Style Sheets)

- CSS is used to **style and design** the entire Student Management System.
- Controls the **colors, fonts, spacing, alignment, background styles, and layout** of each page.
- Makes the application look clean, attractive, and user-friendly.
- CSS ensures **responsiveness**, meaning the website adjusts well on mobile phones, tablets, and desktops.
- Advanced CSS features like flexbox and grid help organize content neatly.

3. JavaScript (JS)

- JavaScript adds **interactivity and functionality** to the Student Management System.
- Handles form validations (e.g., checking empty fields, correct email format, password rules).
- Updates content dynamically without reloading the page—useful for searching students, filtering data, showing real-time attendance updates, etc.
- Used to connect the frontend with backend APIs using **fetch()**.
- Enables features like notifications, calculations (marks/grades), dropdown interactions, and dashboard updates.
- Makes the system more **dynamic, responsive, and user-friendly**.

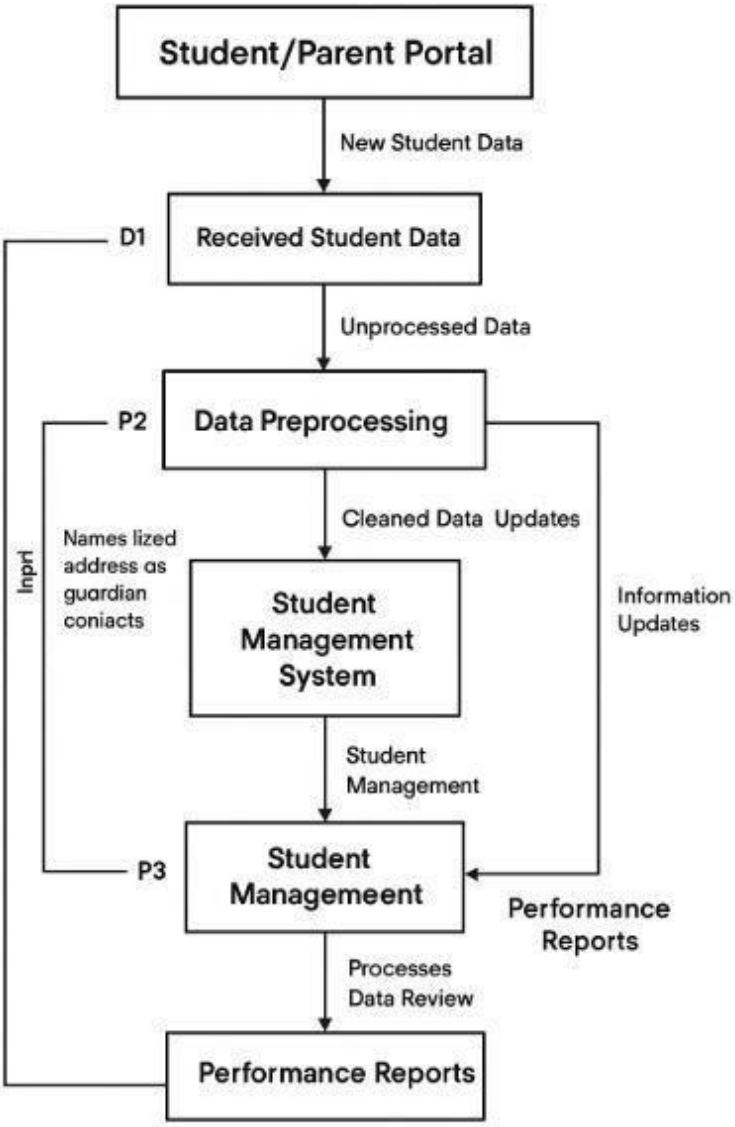
ii. BACKEND TECHNOLOGIES

Node.js (JavaScript Runtime Environment)

- Node.js allows JavaScript to run on the **server side**, not just in the browser.

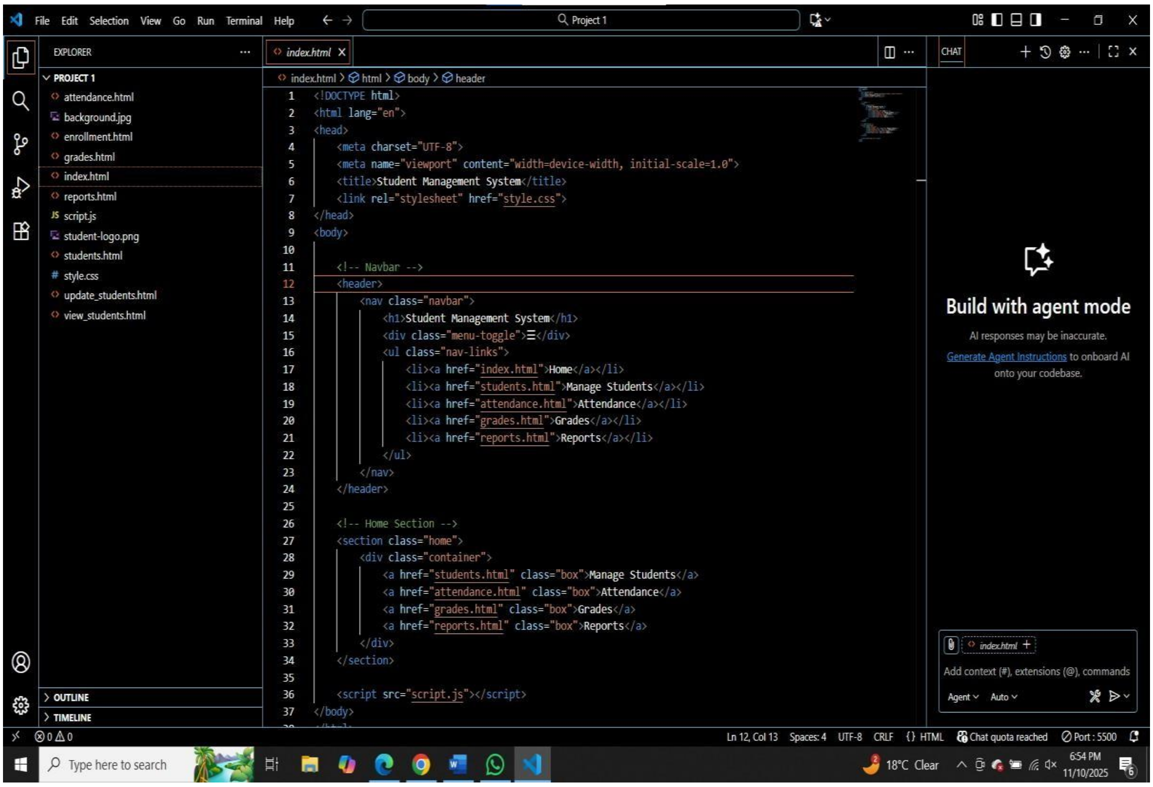
- It is used to create the backend server that handles all background operations like storing data, authenticating users, and processing requests.
- Works on an event-driven, non-blocking architecture, which makes it **fast and efficient** for school management systems.
- Can handle multiple users (admins, teachers, students, parents) at the same time without slowing down

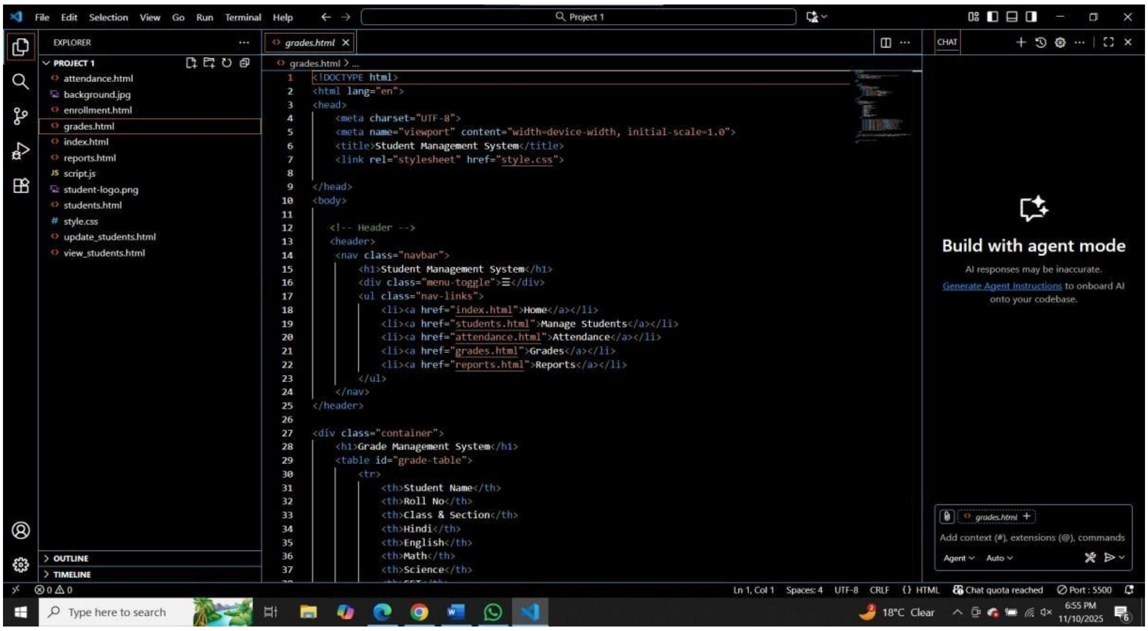
DATA FLOW DIAGRAM

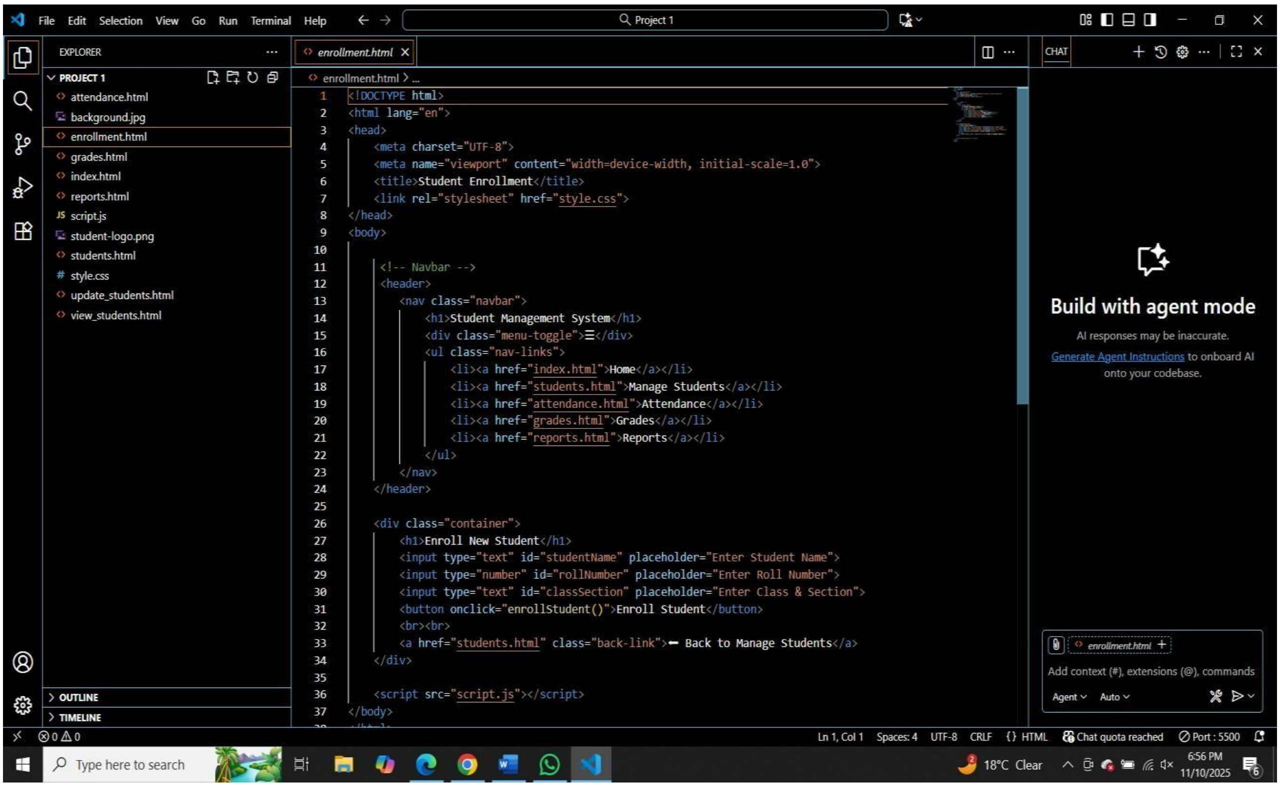


SCREENSHOTS OF PROJECT

CODE SNIPPETS:





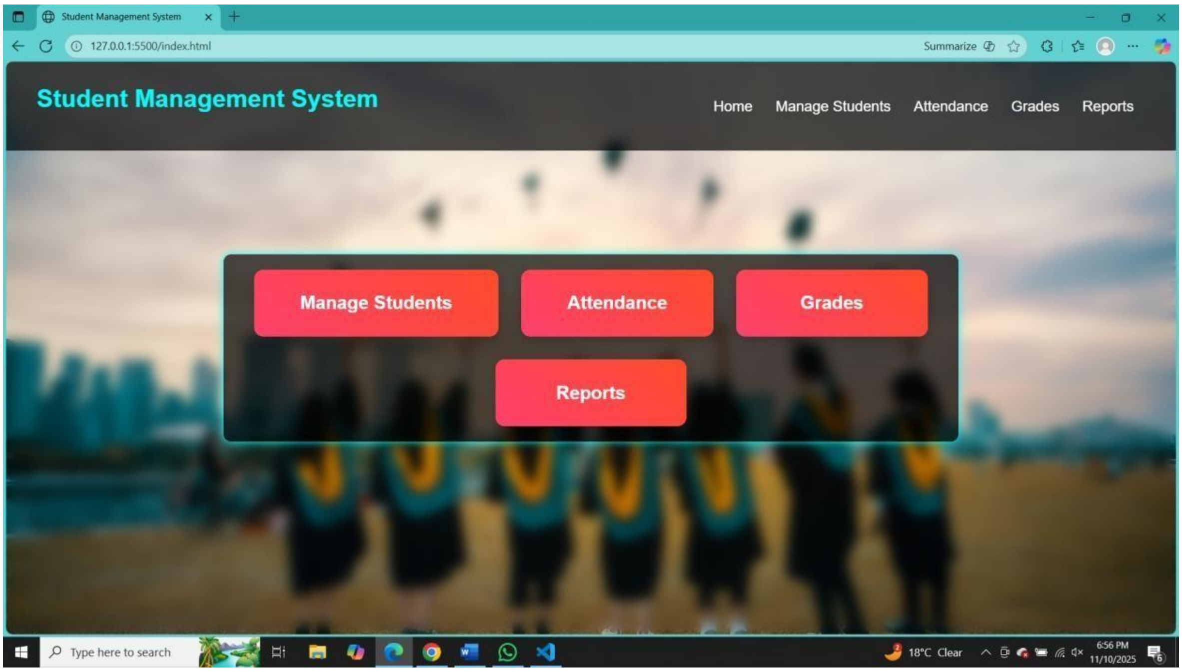


File Edit Selection View Go Run Terminal Help
Project 1
EXPLORER
PROJECT 1
attendance.html
background.jpg
enrollment.html
grades.html
index.html
reports.html
script.js
student-logo.png
students.html
style.css
update_students.html
view_students.html
enrollment.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Enrollment</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<!-- Navbar -->
<header>
<nav class="navbar">
<h1>Student Management System</h1>
<div class="menu-toggle">☰</div>
<ul class="nav-links">
Home
Manage Students
Attendance
Grades
Reports

</nav>
</header>
<div class="container">
<h1>Enroll New Student</h1>
<input type="text" id="studentName" placeholder="Enter Student Name">
<input type="number" id="rollNumber" placeholder="Enter Roll Number">
<input type="text" id="classSection" placeholder="Enter Class & Section">
<button onclick="enrollStudent()">Enroll Student</button>

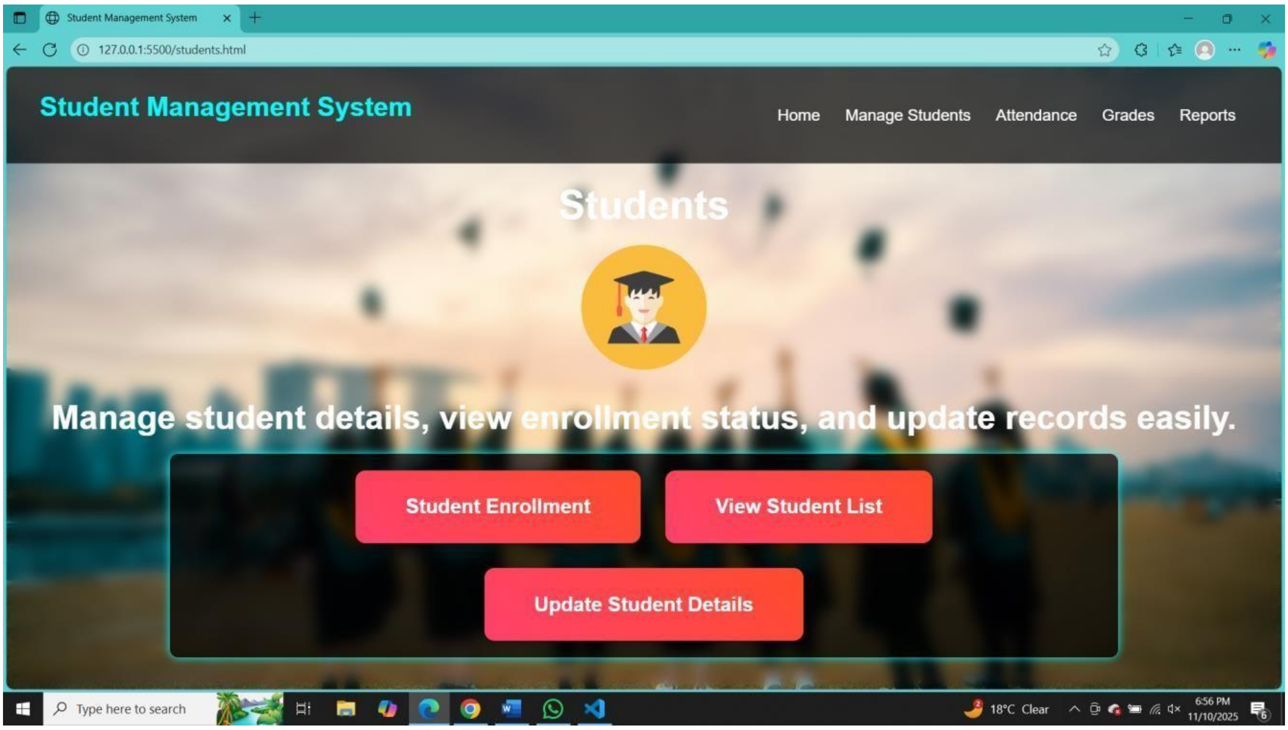
← Back to Manage Students
</div>
<script src="script.js"></script>
</body>
CHAT
Build with agent mode
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
Add context (#), extensions (@), commands
Agent Auto
OUTLINE
TIMELINE
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF HTML Chat quota reached Port : 5500
Type here to search
18°C Clear
6:56 PM
11/10/2025

DASHBOARD

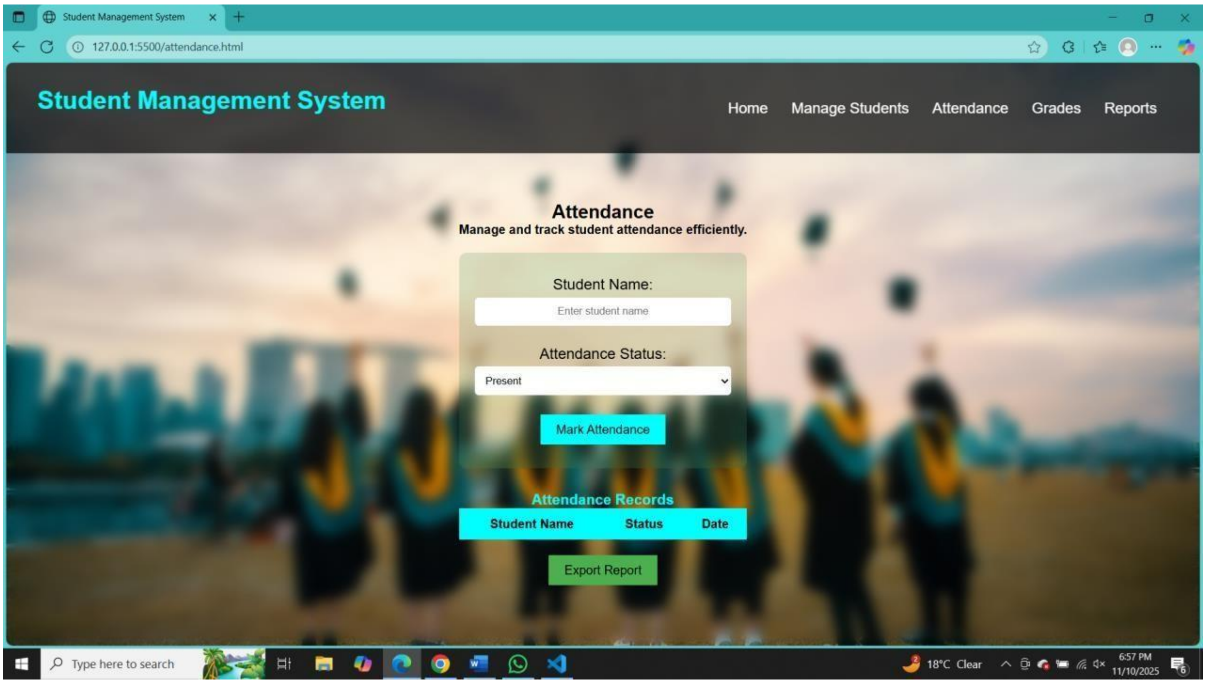


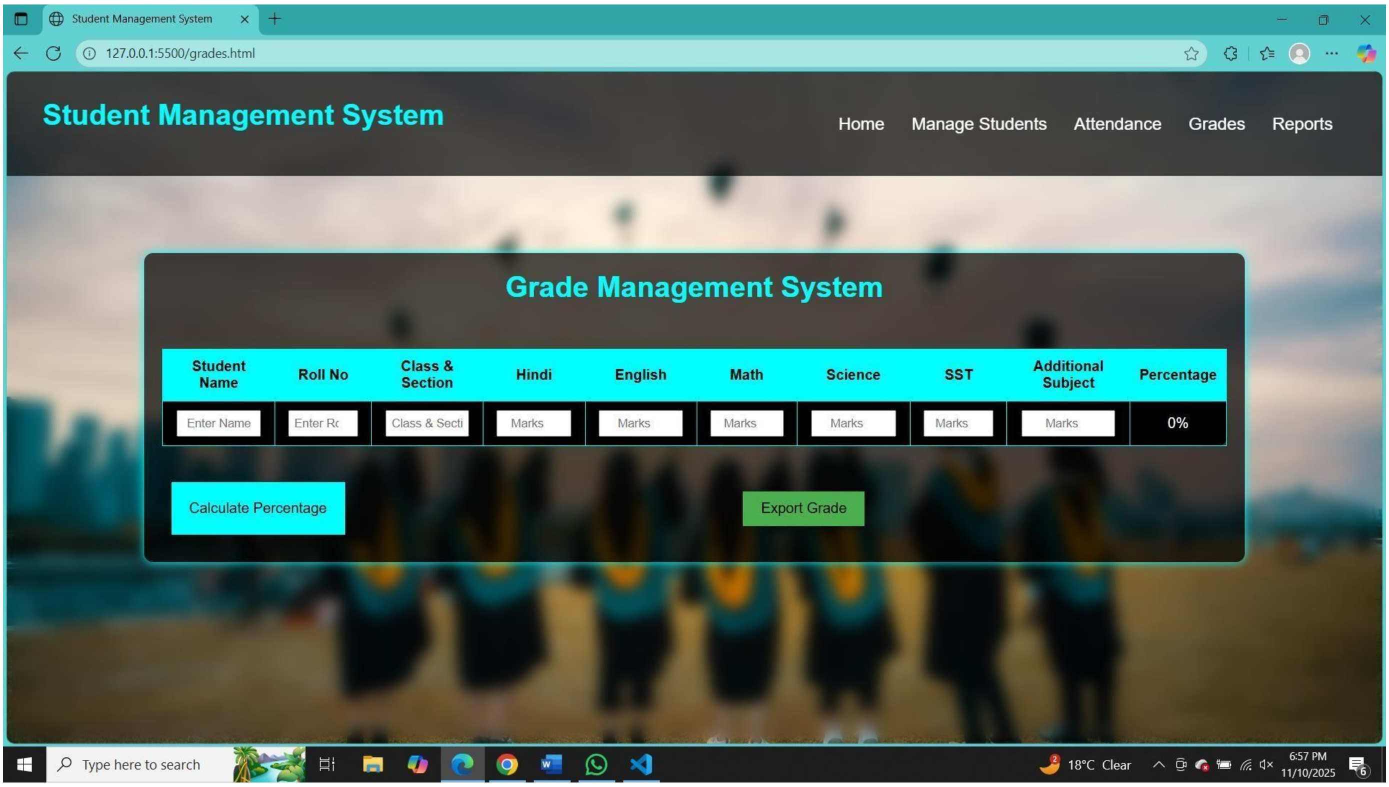
Student Management System
127.0.0.1:5500/index.html
Summarize
Student Management System
Home
Manage Students
Attendance
Grades
Reports
Manage Students
Attendance
Grades
Reports
Type here to search
18°C Clear
6:56 PM
11/10/2025

MANAGE STUDENTS:



AFTER CHANGING LOCATION:




Student Management System
127.0.0.1:5500/grades.html
Student Management System
Home
Manage Students
Attendance
Grades
Reports
Grade Management System
Student Name
Roll No
Class & Section
Hindi
English
Math
Science
SST
Additional Subject
Percentage
Enter Name
Enter Rc
Class & Secti
Marks
0%
Calculate Percentage
Export Grade
Type here to search
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11/10/2025

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion:

The Student Management System successfully streamlines and automates essential academic and administrative tasks within an educational institution. By integrating modules such as student registration, attendance, grades, fees, and reporting, the system minimizes manual work, reduces errors, and enhances overall efficiency. The use of simple and effective technologies like HTML, CSS, JavaScript, and a backend database provides a stable, user-friendly, and scalable platform. The system improves communication between teachers, students, and parents while offering real-time access to important information. Overall, the project demonstrates how digital solutions can simplify operations, support decision-making, and improve the management process within schools or colleges.

6.2 Future Scope:

The system has significant potential for further expansion and enhancement. Future improvements may include integrating advanced features such as biometric or RFID-based attendance, AI-powered analytics for student performance prediction, automated timetable generation, and a full Learning Management System (LMS) for online classes and assignments. Mobile app integration for parents and students can strengthen accessibility and communication. Cloud deployment, multi-school support, and robust data security enhancements can help scale the system for larger institutions. Additional features like hostel management, library management, transport tracking, and fee reminders via WhatsApp or SMS can also be implemented. With these upgrades, the Student Management System can evolve into a comprehensive digital solution for modern educational environments.

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